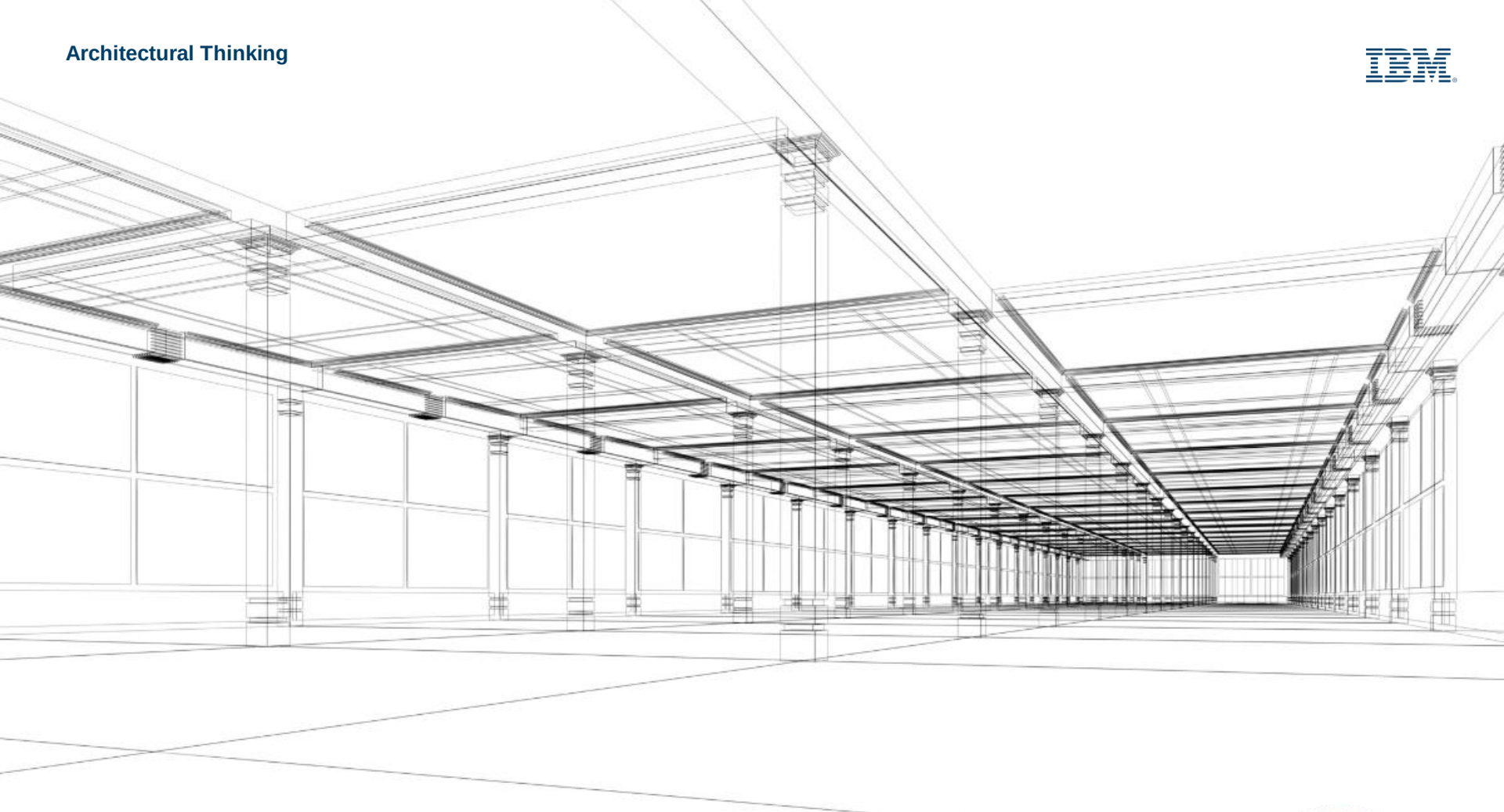




Assets

Version 7.0

Architectural Thinking



Module outline - Introduction and objectives



► Introduction and objectives

Client value and assets

Using assets

Finding assets

Publishing assets

Summary and references



At the end of this module, you should be able to:

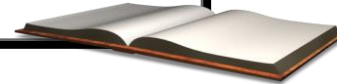
- Understand why architects, IBM and our clients benefit from asset reuse.
- Identify types of architectural assets and explain how they help in solution design.
- Understand how assets are organized in order to reduce the time-to-value for clients.
- Explain when and how to use assets.
- Recognize some key architectural asset community sites.
- Describe why reuse of assets is a good practice.
- Understand the importance of every architect sharing their insights and work at the end of every engagement.



- Let's begin by asking some questions to clarify what we mean when we invoke the term 'asset' and specifically 'architectural asset':
 - Is Wikipedia an asset?
 - Do assets need to be formally approved through a governance process?
 - What is relationship of assets to Knowledge Management?
 - Can an asset become a product?
 - What do we mean by an architectural assets?
 - Where and how do you find architectural assets in IBM?
 - Is the Architectural Overview diagram you developed for your last engagement, but never shared an asset?



Definition: Architectural assets provide a solution to a problem for a given context.



Architectural assets can be either:

- a “*thing*” that represents an idealized or generalized prototypical architectural model - either an individual model element or a sample structure of elements.

Or:

- A “*thing*” that describes “a *way of doing something*” – a technique or approach that is known to reliably achieve some goal.

Note: Many of IBM’s assets are registered as IBM Intellectual Property (IP) with appropriate protections. Some Assets are licensed and these attract substantial fees. If an Asset is of commercial value and offered only to a client for a fee or as part of a commercial service offering, then it may not be available on an open website to all IBMers.

Module outline - Client value and assets



Introduction and objectives

► **Client value and assets**

Using assets

Finding assets

Publishing assets

Summary and references

Stakeholder benefits from architects using assets?



When assets have been used to develop the solution, the benefits to the client can include:

- *Reduced risk*
- *Potential that solution is less ho-hum and more innovative*
- *Potential reduction in solutioning time, cost and delivery*

Client

When assets have been used to develop the solution, the benefits to IBM can include:

- *Reduction in risk*
- *Reduction in cost and potential increase in profit (e.g. time saved may not be passed on to client)*
- *Improved quality and customer satisfaction*
- *Improved communications with delivery team*

IBM

When you can refactor and reuse an existing asset, you do this because:

- *It saves you time*
- *It helps you identify things you may have not considered (i.e. improves quality)*
- *Reduces project risk, when the asset has been proven*
- *Improves your eminence with your colleagues and your client*

IBM Architect



- An asset is only valuable if it reduces the time to value or improves the quality. If it does neither of these then while one might call it an asset, is it really?
- Assets are valuable. They can play the role of “accelerators”; in other words, they do more for less. Assets are also used as risk management mechanisms.
- The time it takes to find an asset is a significant consideration. If it takes more time to locate and use the asset than it would have taken to develop a solution of comparable quality without the asset then there is no derived value in having the asset!

Assets and the organization of assets are a means to the end to reduce the time-to-value for our clients.



What are some of the reasons architects fail to make use of assets:

- Search results in too many/few/wrong assets to investigate – *“sorting the wheat from the chaff just isn’t worth the effort”*
- Time spent re-factoring to meet the client need offsets the value – *“it is quicker to develop it myself”*
- Assets are too fine grained – *“need architectural assets (Principles, ABBs, RAs), not design or development assets”*
- Asset support and currency – *“who created the asset and is the implementation still valid”*
- Limited understanding of quality, usage and constraints – *“I don’t know if it any good?”*
- Assets not created with reuse in mind, does not follow a well-known notation from a standard viewpoint – *“good stuff, but difficult to assimilate and refactor”*
- No obvious mechanism to provide feedback and improve the asset – *“who do I tell if it does not work”*
-
-

Net: Inhibitors across the reuse lifecycle are Cultural, Process, and Technical – it is up to all of us to ‘fix’ these inhibitors





- The term ‘hardened’ assets, refers to an asset that has been specifically developed (perhaps from harvested content) and has undergone a formal review and approval process and (hopefully) has been proven on multiple referenceable projects.
- In contrast a non-hardened asset could be a solution developed by one of your peers and made available without any formal approval process.
 - An example would be wikipedia where ‘assets’ are shared and curated by the community.
- For non-hardened content some sites implement a pre-moderation process, where the asset is reviewed (not necessarily formally approved) before publication and others a post-moderation process, where the asset is reviewed by an assigned moderator after it is published (e.g. GDA+)

Module outline – Using assets



Introduction and objectives

Client value and assets

► **Using assets**

Finding assets

Publishing assets

Summary and references

Using assets in Solution Design



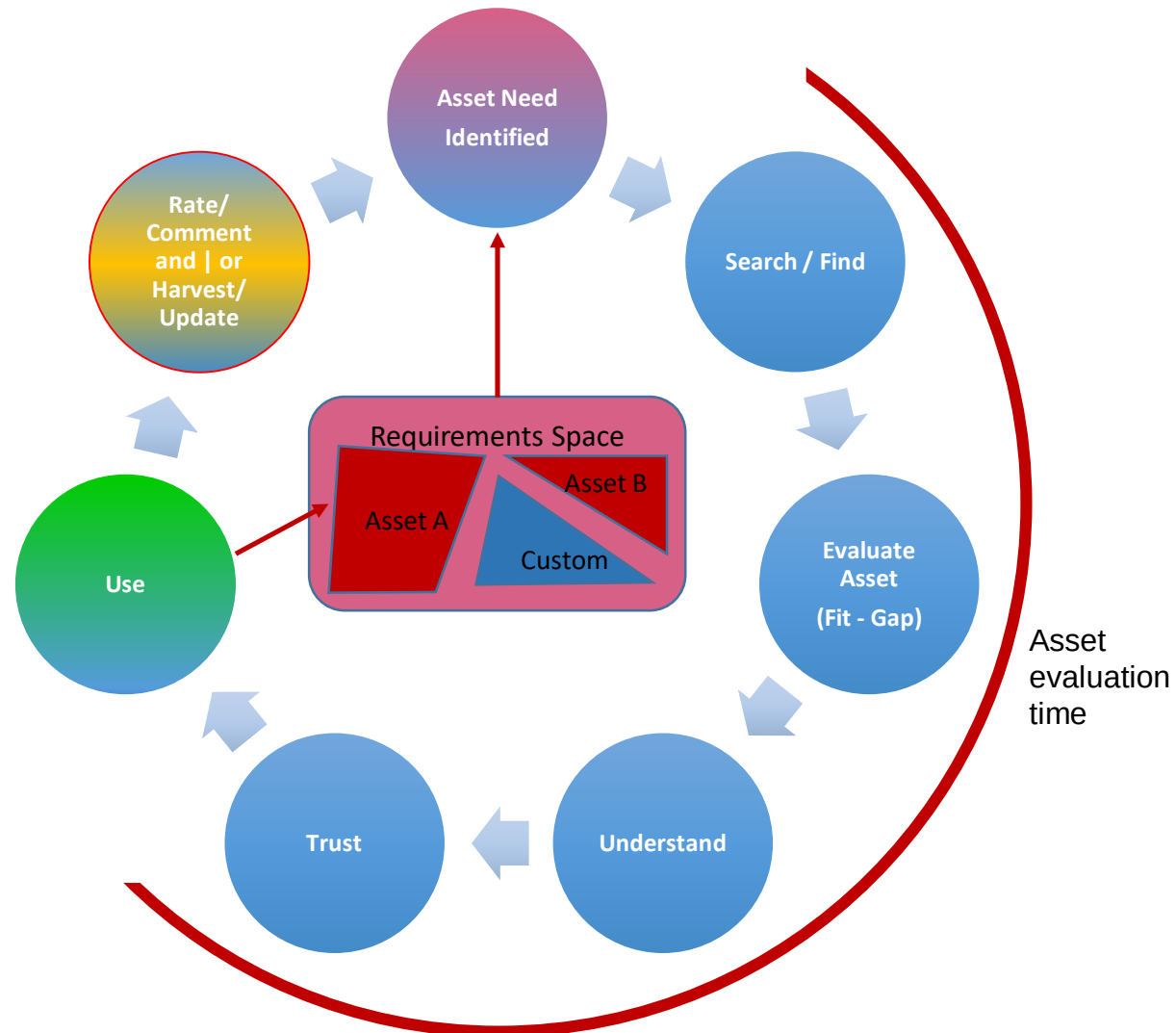
TEAM SOLUTION DESIGN PHASES		
Plan	Pre-Sale Solution Design	Support Implementation and Confirm Value



The Explore activity is the optimal time to consider relevant assets.

As an Architect, you must consider the use of assets before you have fully shaped the solution. Once you have some idea of the client's business requirements or key use cases, you use this input to search for any asset that can be refactored and reused.

You can also use other elements of the target solution to guide your search. This is where the Taxonomy that we discussed earlier ??? becomes relevant as you will want to search on parameters such as Industry, Solution Type, and Technology.



Module outline – Finding assets



Introduction and objectives

Client value and assets

Using assets

► **Finding assets**

Publishing assets

Summary and references

Where can you find assets?



Before you even start searching IBM sites for candidate assets, you should start from your client's perspective: Are there any assets available at an industry-level, from other providers or within the client's organization that can provide value to your Solution Design?

What value does your client expect from an IBM Solution?

- Think in terms of completeness and connection to market leading technology and relevant industry references.

Who are your IBM colleagues who are dedicated to serving this client and what insights can they provide you?

- Talk to a sales leader who has the best relationship to the Business Sponsor.
- Ask the Client Technical Architect (CTA), GTS Chief Architect, Software Architect, or other IBM Technical Leaders who serve the client for their insight.



Assemble the parameters that you want to use in your search. Analyze to determine which are primary or essential for a match, and which are secondary.

Then, consider the available sites for assets.

- You may also have local assets for your team or practice. (*Note: Avoid the Anti-Pattern of only using what is readily available*)
- And you will, quite naturally, be influenced by a recommendation from a trusted colleague.

The guidance here is:

- Check more than one source. Do as thorough an asset search as is justified in the circumstances. Remember that a positive match has benefits in solution value and our ability to deliver for the client.



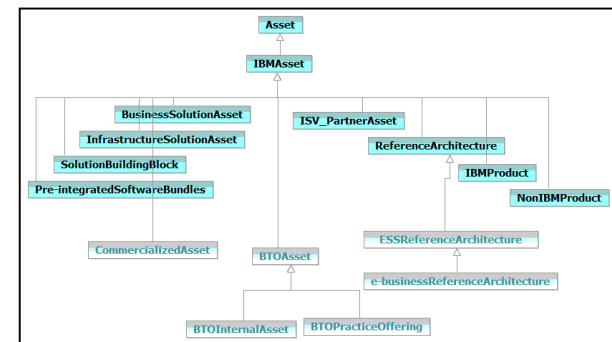
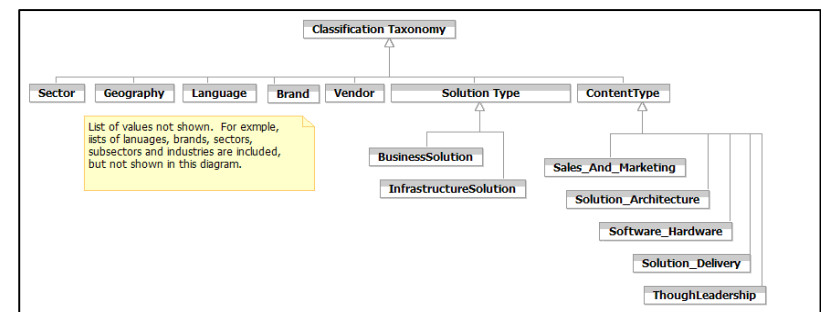
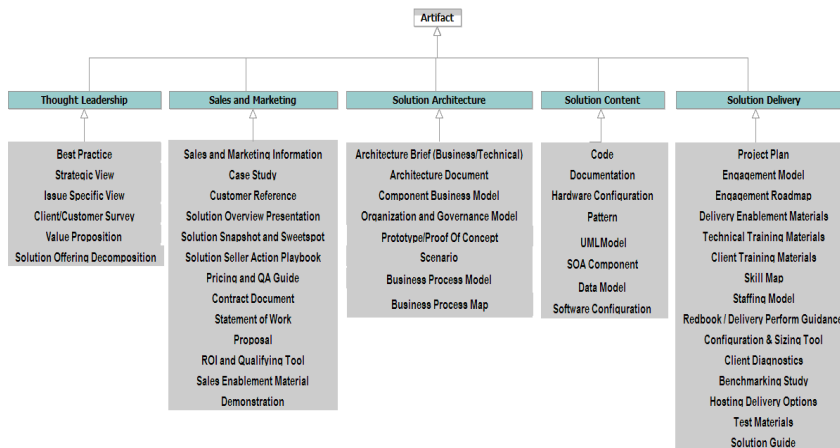
It is helpful to view IBM asset hubs or sites in terms of providing one or more of the following capabilities:

- Asset repository – i.e. file system
 - Metadata describing asset (links to asset in external repository)
 - Search
 - Social
- ~~iRAM is the preferred asset management system in IBM and provides all these capabilities. The underlying platform is IBM Rational Asset Manager. ([link](#))~~
 - There are other key sites architects use to find assets, some more formal than others, such as^(*):
 - ~~Practitioner Portal Lighthouse~~
 - GTS Global Delivery Architecture (GDA+)
 - Solution Gateway.
 - Solution Seeker
 - Technical Collaboration Hubs (Connections Communities)

(*) Note This is far from an exhaustive list



- Understanding how an asset is classified and the metadata describing the asset is key to being able to navigating, uploading and finding assets (This is where the topic of Assets intersects with the discipline of Knowledge Management.)
- The IBM Asset Architecture Board (AAB) has provided insight on both Structural Taxonomy and Classification Taxonomy for assets, primarily used in iRAM.

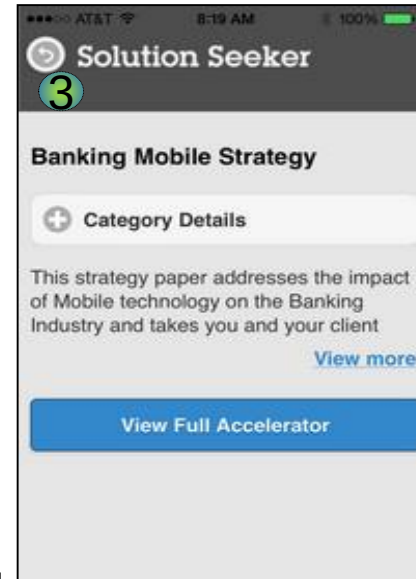
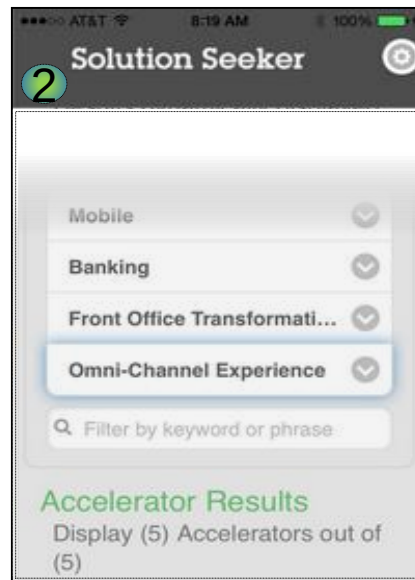
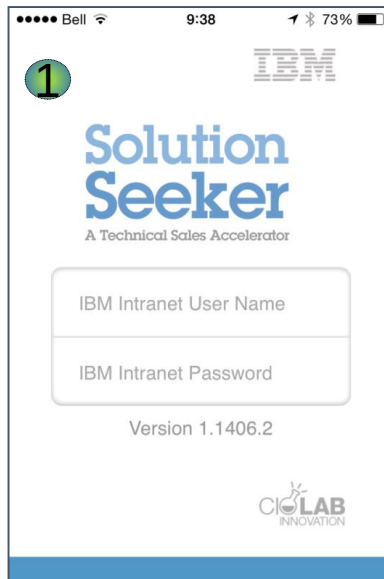


- There are other taxonomies from an industry or business unit perspective that help classify and locate assets (e.g. the GTS GDA+ three tier taxonomy is based on ITIL and TOGAF to classify infrastructure architecture assets)

Access accelerators from web or mobile



[Solution Seeker](#) is an **Asset Meta Search** engine that uses multiple asset repositories and allows the search to be filtered by Industry, Service Line, or Content Type.

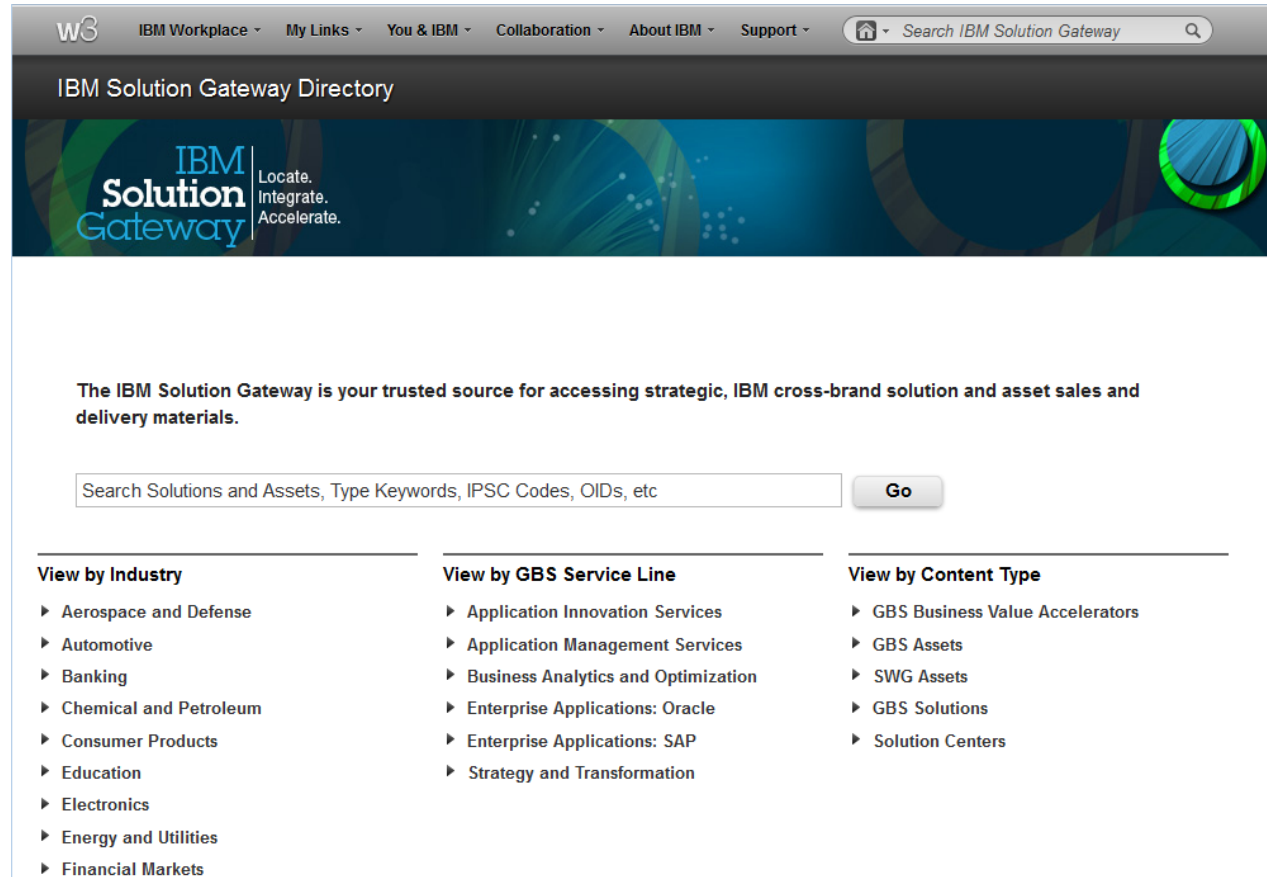


- 1 Log in.
- 2 Select your industry, solution, and your growth plays.
- 3 Review accelerators.



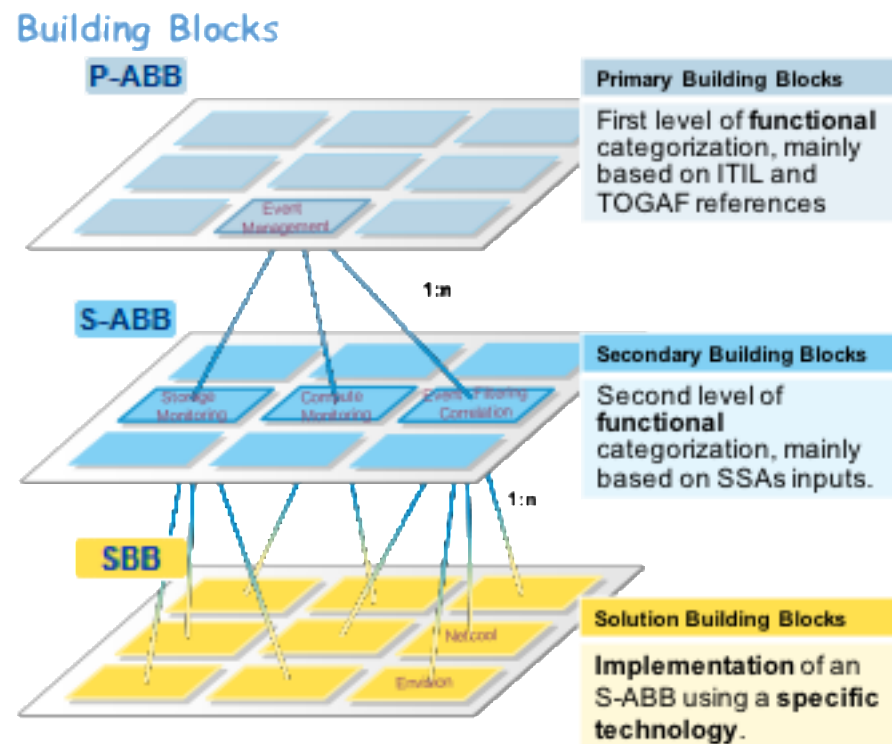
[The IBM Solution Gateway](#) is an **Asset Meta Search** engine that uses multiple asset repositories and allows the search to be filtered by Industry, Service Line, or Content Type.

1. Launch **Solution Gateway**.
2. There are two modes to search:
 - a. Enter a **search string**, or
 - b. Select views by industry or service line to navigate the content.
- A detailed **result view** will be displayed.
3. Choose a result area to open another window and view all relevant information grouped by tabs.





- The GTS [Global Delivery Architecture \(GDA+\)](#) is portal specifically for GTS architects which classifies content based on a three tier taxonomy, where the top tier Primary Architecture Building Blocks (ABB) is based on ITIL and TOGAF. The second tier Secondary Building Blocks is technology non-specific (e.g. RAs) and the third tier, Solution Building Blocks (SBB) are real technology solutions containing TeamSD Work Products.
- The GDA+ approach is to be an open community where any member of the community can add or edit SBB's
- GDA+ changes are post-moderated, meaning that everyone can see and edit immediately any change is made.
- The platform is social allowing members to rate, comment and share content.
- Existing solutions can be searched via the framework or through search.



Technical Collaboration Hub provides insight on Growth Play Assets



▼ Technical Collaboration Hub

Overview

Recent Updates

Status Updates

Members

Forums

Bookmarks

Files

Wiki

▼ Subcommunities

■ Analytics Technical Collaboration Hub

■ Cloud Technical Collaboration Hub

■ Cloud Experts Council

▼ Bluegroup

▼ Community Description

Technical Collaboration Hub

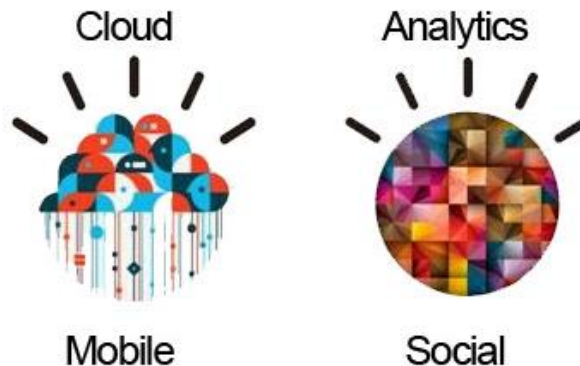
A Cross-IBM Initiative to Transform Technical Sales

Welcome to Technical Collaboration Hub (TCH) community.

CLOUD, ANALYTICS, MOBILE, and SOCIAL are the four key growth strategies for our clients & IBM. IBM's Technical collaboration Hub is the place where IBMers can find the best assets and the experts in these four strategic areas. The hub enables technical solution architects to engage the best "Subject Matter Experts", find the best solution assets & education, take expert opinion on solutions.

Services provided by Technical Collaboration Hub

- Engage with Global Experts across IBM.
- Get your solution reviewed by Experts Council.
- Leverage Assets for Solution Design



Quick Links

[TCH Engagement Form](#)[Cloud TCH](#)[Cloud Experts Council](#)[Analytics TCH](#)[Analytics Experts Council](#)[IBM Expertise Locator](#)[Client Collaboration Hub](#)[Blue Mine](#)



- The [Lighthouse](#) is an **Asset Meta Search** engine that uses multiple asset repositories and allows the search to be filtered by Industry, Service Line, or Content Type.

The screenshot shows the IBM Practitioner Portal Lighthouse interface. At the top, the header reads "IBM Practitioner Portal". Below this, a large blue banner displays "Knowledge Search Results" with a search bar and a "SAVED SEARCHES" dropdown. A sidebar on the left contains icons for search, knowledge, experts, and content. The main content area shows "about 1,458 results found" and a "YOUR FILTERS" section with options for CATEGORIES, TAGS, RATINGS, and DATE. The results are sorted by "Highest Rating". Three results are visible: "Legacy Modernization Client References (ISV Core Banking...)", "Communications: BSS OSS proposal and Solution Kit for Telecom", and "ITOA Requirements IPD". Each result includes a thumbnail, title, owner, modified date, rating, and action buttons like "ADD TO LIBRARY" and "DOWNLOAD".

IBM Practitioner Portal

Knowledge Search Results

SAVED SEARCHES

TeamSD

GENERAL SOCIAL EXPERTS ANALYST CONTENT

about 1,458 results found

YOUR FILTERS CLEAR ALL

CATEGORIES TAGS RATINGS DATE

Sort BY: Highest Rating

SAVE THIS SEARCH SHARE THESE RESULTS

Legacy Modernization Client References (ISV Core Banking...)

OWNER: PABLO.PADIN@ES.IBM.COM | MODIFIED: 5 NOV 2015

AgendaBusiness Motivation and Goals of this Renovation ProjectProject Overview and Renovation Techniques UsedIBM...

5.0 4

ADD TO LIBRARY DOWNLOAD

Communications: BSS OSS proposal and Solution Kit for Telecom

OWNER: SBERNSTN@US.IBM.COM | MODIFIED:

Download the kit: BSS OSS proposal and Solution Kit for Telecom (IBM intranet id and password required to...

5.0 1

ADD TO LIBRARY

ITOA Requirements IPD

5.0 1

Module outline – Publishing Assets



Introduction and objectives

Client value and assets

Finding assets

Using assets

► **Publishing assets**

Summary and references



- Historically architects have not been uniformly good at sharing. What do you think are some of the reasons for this?
- When you have mastered the art of Asset reuse, then is a good time to consider how you can assess and share your solution architecture work.
- We can easily become too precious about what constitutes an ‘asset’. A good way to think about harvesting and sharing your work is to ask yourself the question:
 - *“do I believe that another practitioner reusing this content would be able to develop a similar solution with less effort than having to start from nothing?”*
 - or to put it another way *“do I believe that had I had this content before I started that I would have been able to use to develop solution with less effort than having to start from nothing?”*
- Having made use of an asset, you can from your experience improve the existing asset, either where possible directly updating (e.g. GDA+) or by contacting the owner.
- Simply rating and commenting on an asset adds value to the next person contemplating using, does not have to be positive.
- A best practice is to team up with other colleagues to help generalize the harvested content for reuse.

We all need to make it part of our culture, to openly share, since we all benefit!

Module outline – Summary



Introduction and objectives

Client value and assets

Finding assets

Using assets

Publishing assets

► **Summary and references**



Good architectures (and architects) leverage assets wherever they can.

Assets are:

- Things that highlight “good ways of doing things.”
 - a. Principles, policies, and guidelines
 - b. Patterns
 - c. Method
- Things that can be used directly.
 - a. Architectural building blocks and RAs
 - b. Frameworks

Architects should always use anti-patterns as a way of avoiding the mistakes others have made.

- Maybe the best anti-pattern is “*start with a clean sheet of paper!*”
- Maybe the best pattern is “*always use assets!*”

Good architects:

- Look for opportunities to harvest assets from projects and proposals. (*Every project should result in new or updated assets being shared*)

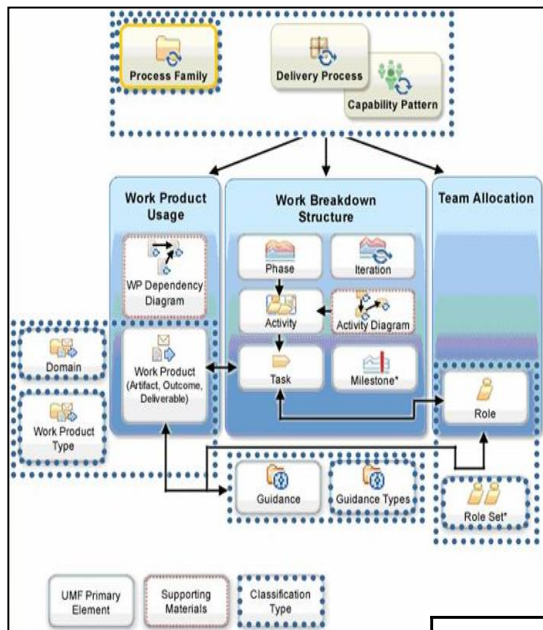


- [IBM Asset Architecture Board Community](#)
- [Lighthouse](#)
- [Global Delivery Architecture \(GDA+\)](#)
- [Solution Gateway](#)
- [Solution Seeker](#)
- Lloyd, PTL and Galambos, GM. *Technical reference architectures*, IBM Systems Journal, [Volume 38, Number 1, 1999](#)
- OMG. *Reusable Asset Specification*, [OMG Final Adopted Specification, ptc/04-06-06](#)
- Shaw, M and Garlan, G. *Software Architecture, Perspectives on an Emerging Discipline*. Prentice-Hall, 1996, ISBN 0131829572

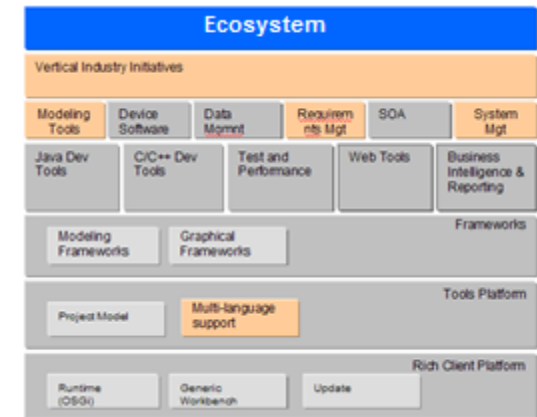
Frameworks (1 of 2)



In the IT architecture context, frameworks provide standard “thinking model” structures that enable communication between Architects. Frameworks, therefore, accelerate the evolution of the Architects’ work.



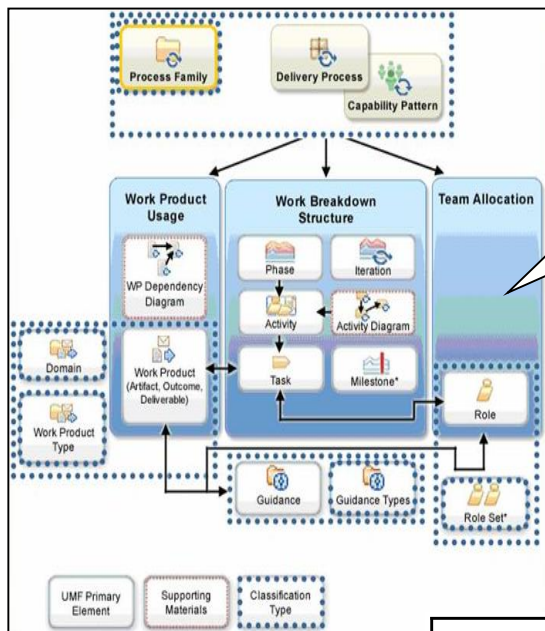
Banking	Chemical & Petroleum	Electronics, Automotive & Aerospace & Defense
Energy & Utilities	Government NCO	Healthcare & Life Sciences
Financial Markets	Insurance	Media & Entertainment
Retail Consumer Products	Travel & Transportation	Telecommunications



Frameworks (2 of 2)

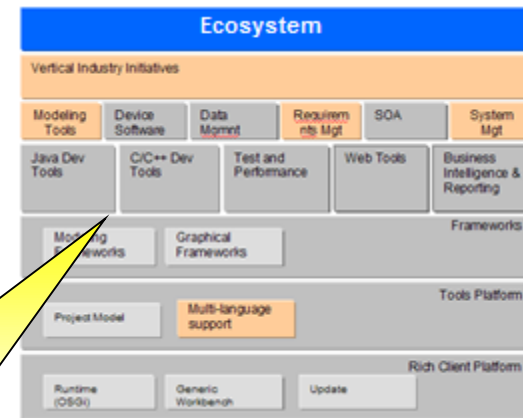


In the IT architecture context, frameworks provide standard “thinking model” structures that enable communication between Architects. Frameworks, therefore, accelerate the evolution of the Architects’ work.



Unified Method Framework: Facilitates understanding and exploitation of a standard approach to projects.

Eclipse Framework: A common platform for integrating and extending tools from different vendors.



IBM Industry Frameworks: Lays out the standard software platforms for IBM industry solutions.

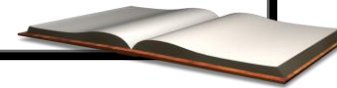
Banking	Chemical & Petroleum	Electronics, Automotive & Aerospace & Defense
Energy & Utilities	Government NCO	Healthcare & Life Sciences
Financial Markets	Insurance	Media & Entertainment
Retail Consumer Products	Travel & Transportation	Telecommunications

What are RAs?



Definition: An architecture that has already been created for a particular domain of interest. It typically includes many different architectural patterns, applied in different areas of its structure.*

* See [LLOYD99]



For the IT architect, there are two types of RA:

- Application-specific (*that is, industry-dependent*)
 - a. IBM's Insurance Application Architecture
 - b. eTOM
- Technology-specific (*often technology vendor-agnostic*)
 - a. IBM Cloud Computing RA
 - b. IBM Mobile RA

Business RAs are assets used to describe businesses, such as IBM's CBM reference models.

RAs should be documented using standard UMF architecture model artifacts.

IBM projects may use:

- IBM's own preferred RAs.
- Those from the client's EA.

Less likely to be from one set of ABBs!

More likely to be from one set of ABBs

See "Positioning and Governing."

Insurance Application Architecture (IAA)



Component Business Model (CBM)

Map view

Category view

Level view

Components

Insurance Application Architecture (IAA)

IAA Foundation Models

Business Terms

Business Model

Business Activity Model

IAA Information Models

Business Solution Templates

Enterprise Model

Data Marts

Operational Data Store

IAA Process & Integration

Enterprise Component Blueprint

Analysis Process Model

Interface Design Model

IAA Insurance Products

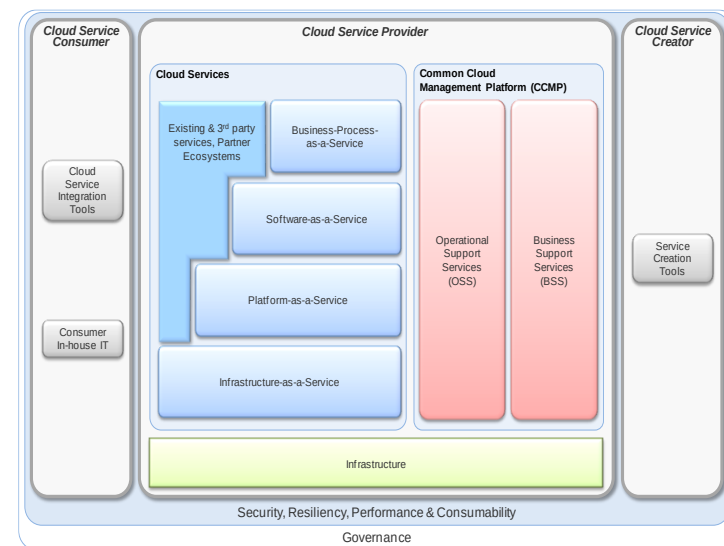
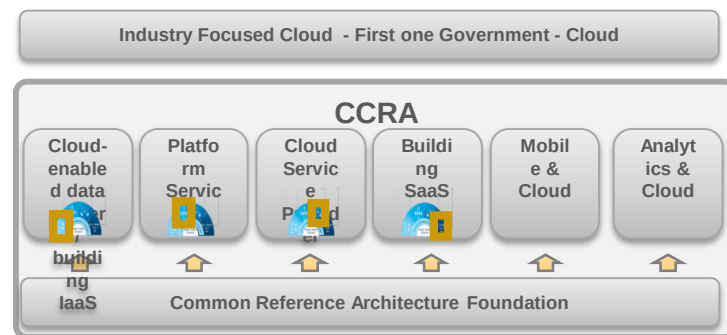
Product Model

Specification Framework

Cloud Computing Reference Architecture (CCRA) 4.0 Overview



- CCRA provides prescriptive guidance on how to build IaaS, PaaS, SaaS, and Service Provider cloud solutions with IBM technology
- CCRA aggregates experience from hundreds of cloud client engagements and IBM-hosted cloud implementations
- CCRA categorize the cloud business models and corresponding architecture by the following “cloud adoption patterns”:
 - Cloud Enabled Data Center (IaaS)
 - Platform-as-a-Service (PaaS) adoption pattern
 - Software-as-a-Service (SaaS)
 - Cloud Service Providers
 - Mobile
 - Analytics
 - Government - Cloud
- For each cloud adoption patterns, CCRA identifies:
 - Common architecture patterns that describe the business drivers, the use-cases, and the technologies that underlie each type of cloud computing implementation.
 - Common architecture patterns for items that cut across all the adoption patterns including security, resiliency, performance, and so on.



When and how are RAs used?



RAs are most valuable in the early phases of the delivery process.

- Perform a fit-gap analysis to understand how the RAs fit your needs and how a decision to not use them might affect the program's TCA and the solution's TCO.

*From
the
outset*

They can also be used for carrying out the architectural assessment of an existing or “under construction” solution.

- Does the solution exhibit “best practice”?
- Does the solution optimize the enterprise's operational effectiveness?

*Forward,
reverse, and
review
engineering*

Use the corresponding artifacts from the selected RAs as a starting point for your solution.

- Do all the “parts” of the RAs come from the same set?
- Which parts require modification?
- Which parts are missing or not needed?

*In an
integrated
way*



THANK YOU



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IBM Rational Asset Manager (iRAM) is the one repository for assets at IBM, with more than 100+ communities. All Business Units (brands) store assets in iRAM.

1. Start the asset search through the **Asset** tab.
2. Enter a **search string**. A detailed **result view** will be displayed.

Narrow down your search by using filters from the left-hand-side navigator.

4. By clicking an asset name in the list, you can open the asset.

Search for Assets

Submit an Asset

Visual Browse

Filter your search

Type

- [Demonstration \[5\]](#)
- [Presentation \[2\]](#)
- [Sales and Marketing Information \[3\]](#)
- [Solution Brief \[2\]](#)
- [Supporting Asset \[19\]](#)
- [View more](#)

Category

- [Conformance and Access \[9\]](#)
- [Geography and Language \[30\]](#)
- [Products, Technologies, and Standards \[5\]](#)
- [Sectors and Industries \[5\]](#)
- [Storage Services \[18\]](#)
- [View more](#)

State

- [Approved \[34\]](#)
- [As is \[1\]](#)
- [Draft \[2\]](#)

Community

- [GTS IT Optimization CoP \[3\]](#)
- [GTS Storage Services CoP \[18\]](#)
- [IBM Industry Transformation Toolkit \[2\]](#)
- [IBM Sensor Solutions \[3\]](#)
- [S and D Business Solution](#)

Search for Assets

smarter planet systems

GO

Filters: none

Advanced

Previous | 1-25 of 37 | Next

Name	Version	State	Community	Rating	Modified	Relevance
T&T - Solutions for a Smarter Planet: Railroad and Airports, Airlines- Presentation	1.0	Approved	S and D Business Solution Center BSC La Gaude	★★★★★	Jun 2, 2010	
Smarter Manufacturing	1.0	Approved	IBM Sensor Solutions	★★★★★	Aug 18, 2010	
Smarter Item Inventory with OAT System	1.0	Approved	IBM Sensor Solutions	★★★★★	Aug 11, 2010	
Smarter Industrial Operations	1.0	Approved	IBM Sensor Solutions	★★★★★	Aug 11, 2010	
E&U - Les visages de Smart Grid dans le monde / Worldwide Faces of Smart Grid- White Paper	1.0	Approved	S and D Business Solution Center BSC La Gaude	★★★★★	Aug 21, 2010	
E2E - Some experiences gained and lessons learned from a major E2E System Design Project	1	Approved	IBM Enterprise Architecture	★★★★★	Sep 23, 2008	